UVU Club Volunteer Scheduling & Hours Tracking System

System & Analysis

A volunteer hours management app for UVU Students and Staff

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Table of Contents

0 – Executive Summary 1

0-1 Introduction 2

0-2 Client Description 2

0-3 Statement of Problem 2

0-4 Target Audience 3

0-5 System Discription 4

0-5.1 System Discription and Capabilities 4

1 – Process Model 5

1-1 Written Use Case for Create a Volunteer Event 5

1-2 Use Case and Swimlane Diagrams for Create a Volunteer Event 6

1-3 Written Use Case for Volunteer Sign Up Event 7

1-4 Use Case and Swimlane Diagrams for Volunteer Sign Up Event 8

2 – Logic Model 10

2-1 Decision Table 10

3 – Conceptual Model 11

3-1 ER Diagram 11

3-2 Data Dictionary 13

4 – Interface Design 15

4-1 Web Interface Leader 15

4-2 Home and Dashboard 16

4-3 Event Management and Reports 17

4-4 Mobile Interface 18

4-5 Help and Error Messages 19

5 – Appendix Material 20

List of Tables & Table of Figures

List of Tables

Table 1	Pages 6
Table 2	Pages 8
Table 3	Pages 10
Table 4	Pages 13-14

Table of Figures

Figure 1	Pages 7
Figure 2	Pages 9
Figure 3	Pages 12
Figure 4	Pages 15
Figure 5	Pages 16
Figure 6	Pages 17
Figure 7	Pages 17
Figure 8	Pages 18
Figure 9	Pages 19

0 Executive Summary

The Volunteer Scheduling and Hours Tracking System is designed to be efficient on how UVU student clubs manage events, sign-ups, and administers volunteer hour reporting. Currently, clubs rely on either paper sheets, group chats, or spreadsheets, which leads to inconsistent data, possibility of lost attendance records, and extra administrative work for club leaders and UVU administrators. This inconsistency also impacts UVU students and volunteers by risking their scholarship opportunities due to lost hours. Our system addresses these concerns by integrating event creation forms, volunteer sign-ups, attendance tracking, and even tracking hours into one centralized application. Through the process models developed using Use Cases, Swimlane diagrams, ER Diagrams, Data Dictionaries, and Interface Design (web and mobile) will present how club leaders and volunteers interact with the system to perform the essential tasks.

0-1 Introduction

0-2 Client Description

Utah Valley University hosts numerous student clubs that coordinate service and volunteer activities throughout each term. Club structures are typically straightforward: a president and officers plan initiatives while members respond to announcements via group messaging, email lists, and campus postings. Faculty advisors provide oversight, and many clubs submit end-of-term activity summaries to the UVU Clubs Office

Current processes rely on a mix of spreadsheets, informal chat threads, and occasional paper sign-ups. This ineffective method contributes to capacity mismatches, incomplete attendance confirmation, and manual reconciliations for hour totals. A centralized, mobile-friendly scheduling system would standardize event listings with slot limits, streamline sign-ups, and support hour approvals, and produce cleaner reports for admins and the Clubs Office.

0-3 Statement of Problem

Currently, UVU clubs rely heavily on manual methods to manage volunteer sign-ups and track service hours. Leaders often use paper sign-in sheets, informal group chats, or ad-hoc spreadsheets, which creates inconsistencies and a high risk of losing records. Volunteers may forget to sign in, or hours may not be accurately transferred into a central record, making it difficult for coordinators to verify attendence. As a result, clubs face administrative overhead,

reduced accountability, and frustration among members who want reliable recognition for their contributions.

This project presents an opportunity to streamline and modernize the process. By creating a dedicated volunteer scheduling and hours tracking system, UVU clubs will be able to centralize event management, reduce manual errors, and give both leaders and members a clearer picture of participation. Automating these processes will not only improve efficiency but also enhance transparency, ensuring service hours are recorded accurately and can be reported for scholarships, leadership positions, and campus engagement metrics.

0-4 Target Audience

The primary audience includes UVU student club leaders and coordinators, who are responsible for organizing events, assigning volunteers, and reporting service hours to the university. They need a system that is simple, reliable, and accessible on demand.

The secondary audience is student volunteers across various clubs. These students often balance academics, part-time work, and extracurricular activities, so they need an intuitive platform that allows them to sign up for events quickly, track their hours, and view their participation history. Finally, university staff or advisors who oversee club operations form a tertiary audience. They will use the system to audit hours, ensure compliance with UVU guidelines, and provide oversight on club activities.

0-5 System Description

The proposed system will provide a centralized web application for creating events, allowing volunteers to sign up, and automatically logging service hours once participation is confirmed. Club leaders will be able to manage events, track attendance, and export reports, while students can view their past contributions. The design will prioritize a clean, user-friendly interface that minimizes time spent on administrative tasks and maximizes accuracy.

0-5.1 System Description and Capabilities

- Event creation and scheduling by club leaders
- Volunteer sign-up and confirmation features
- Automatic logging of hours based on participation
- Reporting and exporting tools for leaders and coordinators
- Volunteer dashboards to view total hours and upcoming events
- Administrative oversight functions for faculty or staff

Summary: Together, these capabilities provide a full life-cycle solution for managing club volunteers—from event creation through hour verification—reducing administrative burden and giving students confidence in the recognition of their service.

1 Process Model

The process models will demonstrate how the Club Volunteer Scheduling System supports two important functions: Creating volunteer events and allowing volunteers to sign up for them. These models will display the step by step logic of actions between the users and the system, to show how decisions are being made at each stage. The *Create Volunteer Event* process focuses on how the club leaders will input and manage event details. The *Volunteer Sign-Up* process focuses on how students register for available volunteer opportunities. Each process model helps visualize and provides a clear understanding of the system logic and user interactions.

1-1 Written Use Case for Create a Volunteer Event

The Create Volunteer Event Use Case describes how a club leader interacts with the system to post new volunteer opportunities for their members/volunteers. It first begins with the Club Leader logs in and selects the “Create Event” option from the Dashboard. The system then displays a form where the Club Leader can enter the event details (Event Name, Date, Time, Location, and number of participants.) Once the Club Leader submits the event information, the system verifies the inputs. If the information of the event passes the validation checks, it is saved to the database and becomes available to volunteers in the event listings. Overall, this use case helps Club Leaders establish better communication and participation with their members by effectively organizing and publishing events in the same system.

1-2 Use Case and Swimlane Diagrams for Create a Volunteer Event

The use case diagram below identifies the club leader's role and outlines the functions of entering event details and submitting them for validation. The swimlane diagram provides a visual representation of each logical step in the workflow.

Use Case Title: Create Volunteer Event
Primary Actor: Club Leader
Level: User Goal (Blue)
Stakeholders: Club Leaders, Volunteers, UVU Club Office
Precondition: Club Leaders sign in with their credentials and are authorized to create events.
Minimal Guarantee: The Event details are saved as a draft when not completed.
Success Guarantee: The new Event is successfully created, stored in the database, and available to Volunteers.
Trigger: Club Leaders selects "Create Event" in the Club Volunteer Scheduling System.
Main Success Scenario: 1. Club Leader selects "Create Event" from the Dashboard. 2. System displays the event creation form. 3. Club Leader enters event information (event title, date, time, location, number of participants, description of event.) 4. System validates the information entered on the creation form. 5. If all required inputs are completed, the system saves the new event. 6. System confirms the event has been created and displays it on the list of Volunteer Opportunities dashboard. 7. Volunteers have access to view the event and sign up.
Extensions: • Missing Required Field in creation Form: System alerts the Club Leader to complete the missing required field when they try to submit the form. • Invalid Date/Time: System alerts the Club Leader and requests valid format (mm/dd/yy, Time.) • System Database Error: System shows "Unable to save event. Please try again."

Table 1 Use Case for Create a Volunteer Event

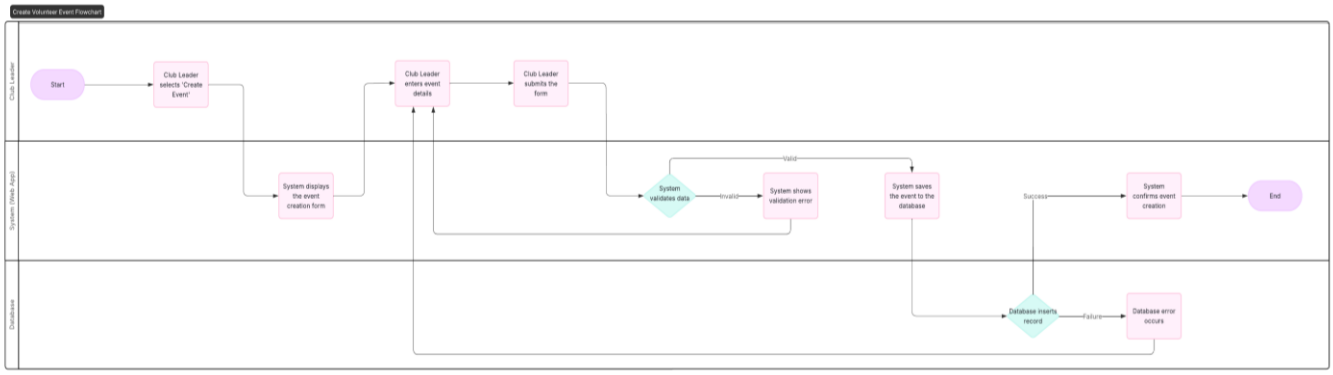


Figure 1 Swim Lane Diagram for Create a Volunteer Event

1-3 Written Use Case for Volunteer Sign Up Event

The Sign Up for Event Use Case describes how a volunteer registers for a club volunteer opportunity through the application system. Once the volunteer is logged in, they will browse through the available events and select one to view its details (date, time, location, and remaining number of participants.) When the volunteer selects “Sign Up”, the system will validate the availability and checks the volunteer’s existing commitments for scheduling conflicts. If there are no scheduling conflicts and there is an open spot for the event, the system will confirm the registration and update the registered list. If the event is full and user tries to sign up, there will be an error message. This use case helps simplify the sign-up experience for volunteers while maintaining accurate attendance records for the Club Leaders.

1-4 Use Case and Swimlane Diagrams for Volunteer Sign Up Event

The use case diagram below shows the outline functions from the perspective of the volunteer's interaction with the event listings within the system and the goal of signing up for opportunities. The swimlane diagram below provides visual details on the logical process the system follows.

Use Case Title: Sign up for Event
Primary Actor: Volunteer
Level: User Goal (Blue)
Stakeholders: Volunteers, Club Leaders, Club Advisors, UVU Club Office
Precondition: 1. The Volunteer is logged into the system. 2. The event is published in the system.
Minimal Guarantee: The system records the volunteer's registration, and displays an error message if it conflicts with the event being full or existing commitments.
Success Guarantee: Volunteer is successfully registered and the number of participants in the event is updated.
Trigger: Volunteer selects "Sign Up" on the event details page.
Main Success Scenario: 1. Volunteer opens the event details page from the dashboard and selects "Sign Up." 2. System will verify that the sign-up window is still open. 3. System checks the number of available slots in the event. 4. System will check for time conflicts with the volunteer's existing commitments. 5. System records the volunteer's registration and updates the event registration. 6. System will display a success message and will be added to Volunteer's My Events.
Extensions: • Sign Up for Event closed: System displays "Sign up is closed for this event." • Available slots for Event are full: System displays "This event is full." • Conflict on existing commitments: System displays "You are already registered for another event at this time."

Table 2 Use Case for Sign Up for Event

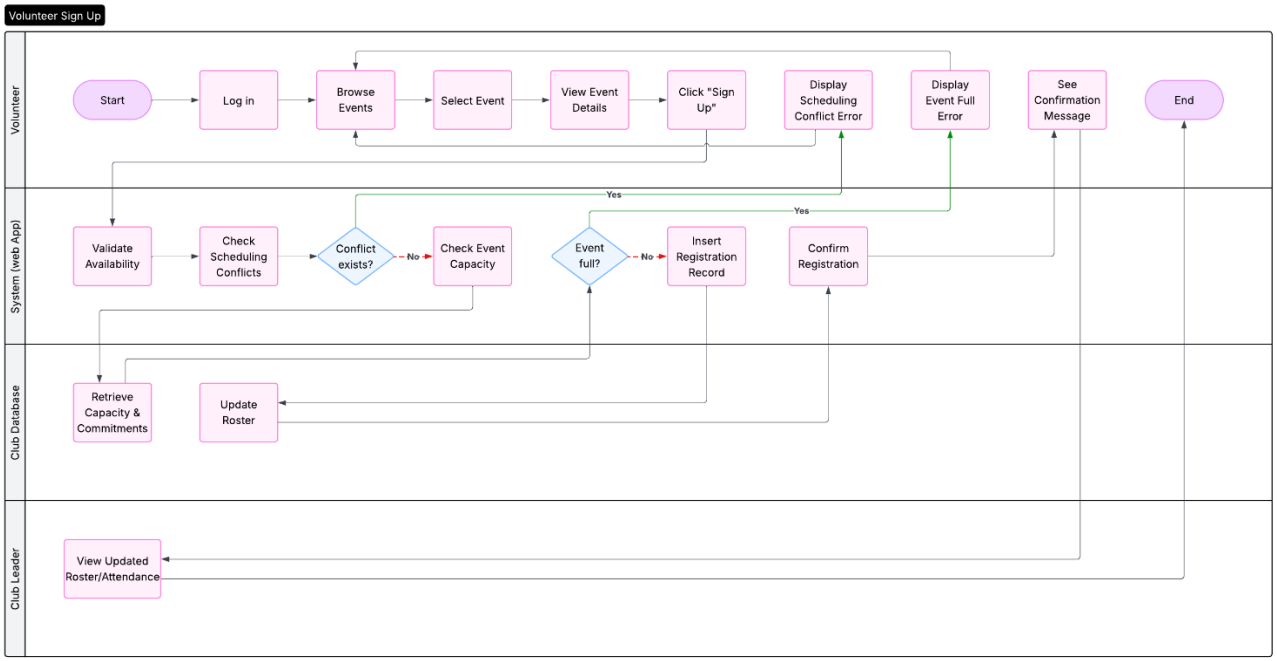


Figure 2 Swim Lane Diagram for Volunteer Sign Up Event

2 Logic Model

The decision table below outlines the logic used to evaluate and approve volunteer service hours within the UVU Club Volunteer Scheduling System. The process considers five key conditions: whether the volunteer was signed up for the shift, whether valid check-in and check-out times were recorded, whether the reported hours fall within the scheduled shift duration, and whether a club leader flagged any issues.

2-1 Decision Table

Based on these conditions in the Decision Table, the system determines the appropriate action: automatically approving the hours when all requirements are met, routing the entry to a leader for manual review when discrepancies or concerns arise, requesting corrections from the volunteer when data is missing, or rejecting the submission entirely if the volunteer was not legitimately signed up. This structured logic ensures consistency, reduces manual workload for club leaders, and improves the accuracy and integrity of volunteer hour reporting.

Rule	C1 Signed Up?	C2 Check -in?	C3 Check -out?	C4 Hours Valid?	C5 Leader Flag?	A1 Auto Approve	A2 Manual Review	A3 Reject	A4 Request Fix	A5 Lock Record
R1	Y	Y	Y	Y	Y	X				X
R2	Y	Y	Y	N	N		X			
R3	Y	Y	Y	Y	Y		X			
R4	Y	N	N	-	-				X	
R5	Y	Y	N	-	-				X	
R6	N	-	-	-	-			X		

Table 3 Decision Table of the system logic behind verification of input information

3 Conceptual Model

The conceptual model of the system is shown by the Entity Relationship Diagram and the Data Dictionary. Both models provide a visual representation of how the information is stored about the users, events, hours, etc. and the logical workflow of the system.

3-1 ER Diagram

The Entity–Relationship Diagram (ERD) shows how the core components of the UVU Club Volunteer Scheduling System relate to one another. At the center of the system is the *User* entity, which represents volunteers, club leaders, advisors, and administrators. Users can belong to one or more clubs, and this many-to-many relationship is managed through the *ClubMembership* entity. Each club may also have an assigned advisor, which is represented by a foreign key relationship from Club to User.

Clubs can create events, and each event is broken into one or more shifts. These entities form the structural framework of the system and define when and where volunteer opportunities occur. The *Shift* entity stores the specific times and volunteer requirements associated with each event. Volunteer attendance is recorded through the *ShiftSignup* entity, which tracks when a user signs up for a shift. Attendance at the event is then documented in the *Attendance* entity, capturing check-in and check-out times along with attendance status. This allows the system to determine whether the volunteer showed up, arrived late, left early, or was absent.

The *HoursLog* entity completes the workflow by storing the calculated hours, any adjustments, and the approval decision made by a club leader or advisor. This entity ties directly to Attendance and ensures that volunteer hours are reviewed and verified before being counted.

Together, these relationships create a clear progression from users and clubs to events and shifts, to volunteer sign-ups, attendance tracking, and final hour approval. The ERD reflects the full lifecycle of volunteer activity within the system and supports accurate reporting and administration.

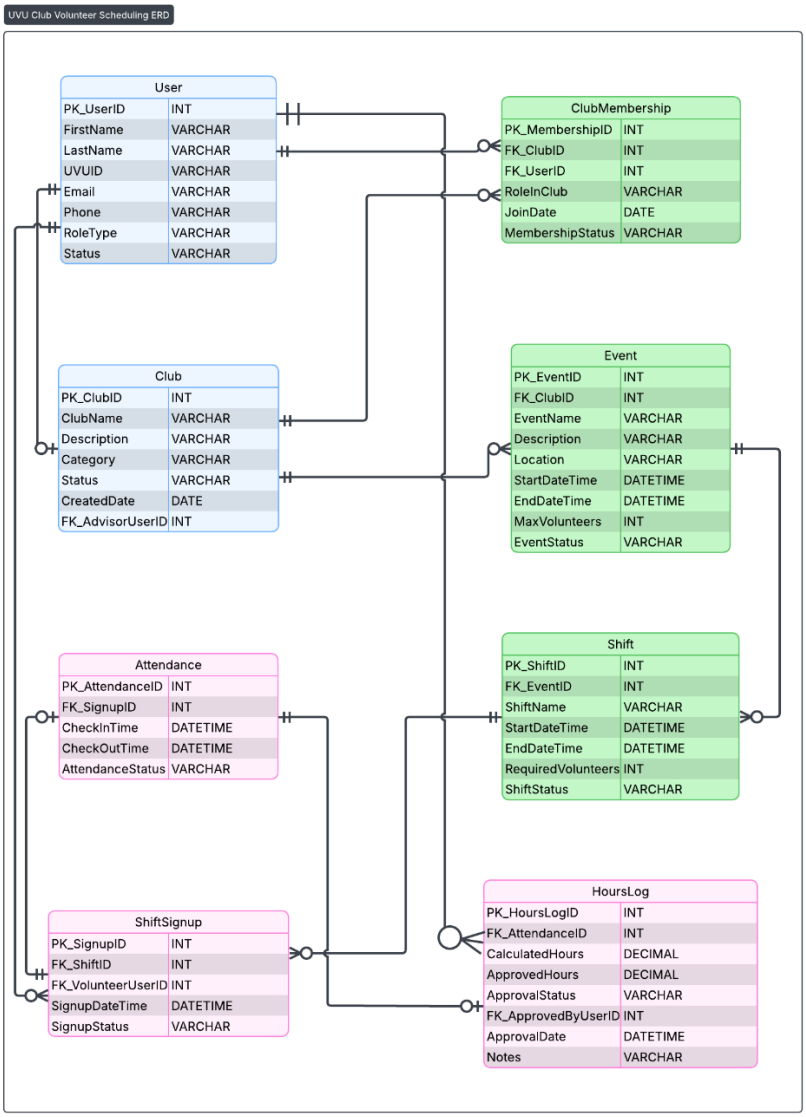


Figure 3 ER Diagram of system

3-2 Data Dictionary

The data dictionary describes how all of the key tables, attributes, and relationships fit into the system's workflow. It defines every field, including data types, constraints, and weather values are required. It also shows how foreign keys link tables together for accurate tracking of clubs, events, shifts, and volunteer activity. This verifies consistency across the system by standardizing names and structures for all its database components. The data dictionary table provides a shared reference for stakeholders and developers for how data is stored and used.

Key	Column	Data Type	Description
User Table			
PK	UserID	INT	Unique identifier for each user
	FirstName	VARCHAR(255)	User's first name
	LastName	VARCHAR(255)	User's last name
	UVUID	VARCHAR(20)	University-issued ID number
	Email	VARCHAR(255)	Unique email address
	Phone	VARCHAR(20)	User mobile phone (optional)
	RoleType	VARCHAR(50)	Volunteer, Leader, Advisor, or Admin
	Status	VARCHAR(20)	Active or Inactive
Club Table			
PK	ClubID	INT	Unique identifier for each club
	ClubName	VARCHAR(255)	Official club name
	Description	VARCHAR(500)	Summary of club purpose
	Category	VARCHAR(100)	Club category/type
	Status	VARCHAR(20)	Active or Inactive
	CreatedDate	DATE	Date club was added
FK	AdvisorUserID	INT	References User.UserID
Club Membership Table			
PK	MembershipID	INT	Unique membership identifier
FK	ClubID	INT	References Club.ClubID
FK	UserID	INT	References User.UserID
	RoleInClub	VARCHAR(50)	Volunteer, Officer, or Leader
	JoinDate	DATE	Date user joined the club
	MembershipStatus	VARCHAR(20)	Active or Inactive
Event Table			
PK	EventID	INT	Unique identifier for an event

FK	ClubID	INT	References Club.ClubID
	EventName	VARCHAR(255)	Name of the event
	Description	VARCHAR(500)	Details of the event
	Location	VARCHAR(255)	Event location
	StartDateTime	DATETIME	Event start time
	EndDateTime	DATETIME	Event end time
	MaxVolunteers	INT	Maximum volunteers allowed
	EventStatus	VARCHAR(50)	Draft, Open, Closed, Cancelled
Shift Table			
PK	ShiftID	INT	Unique identifier for a shift
FK	EventID	INT	References Event.EventID
	ShiftName	VARCHAR(255)	Optional label (e.g., 'Morning Shift')
	StartDateTime	DATETIME	Shift start time
	EndDateTime	DATETIME	Shift end time
	RequiredVolunteers	INT	Number of volunteers needed
	ShiftStatus	VARCHAR(50)	Open, Full, Completed, Cancelled
ShiftSignup Table			
PK	SignupID	INT	Unique signup identifier
FK	ShiftID	INT	References Shift.ShiftID
FK	VolunteerUserID	INT	References User.UserID
	SignupDateTime	DATETIME	When the volunteer signed up
	SignupStatus	VARCHAR(50)	SignedUp, Waitlisted, Cancelled
Attendance Table			
PK	AttendanceID	INT	Unique attendance record
FK	SignupID	INT	References ShiftSignup.SignupID
	CheckInTime	DATETIME	Time volunteer checked in
	CheckOutTime	DATETIME	Time volunteer checked out
	AttendanceStatus	VARCHAR(50)	Present, NoShow, LeftEarly
HoursLog Table			
PK	HoursLogID	INT	Unique hours log
FK	AttendanceID	INT	References Attendance.AttendanceID
	CalculatedHours	DECIMAL(5,2)	Hours calculated from timestamps
	ApprovedHours	DECIMAL(5,2)	Hours approved by leader
	ApprovalStatus	VARCHAR(50)	Pending, Approved, Rejected
FK	ApprovedByUserID	INT	References User.UserID
	ApprovalDate	DATETIME	When approval decision was made
	Notes	VARCHAR(500)	Optional notes for adjustments

Table 4 Data Dictionary for system

4 Interface Design

This system is a web application that also works well on smartphones. Club leaders will mostly use the web version when they plan events and run reports. Student volunteers will usually use the mobile version to sign up, see upcoming events, and check their hours. The main goal is a clean, simple layout for clubs to use without a lot of training and volunteers can utilize the application with ease on either the web or on their smartphone.

4-1 Web Interface

On the web, the page has a header at the top with the system name and menu. The main content sits underneath. Leaders see menu links like Home, Events, Approvals, and Reports. Volunteers who sign in on the web see a smaller menu that focuses on Events and My Hours. It shows the title “UVU Club Volunteer Scheduling and Hours Tracking,” one short line of text, and two fields labeled Email and Password. Below that there is a green Sign In button and a small “Forgot password?” hyperlink. The whole login section is centered for a modern simplistic style to look the same on various screen sizes.



Figure 4 Prototype of Log In page for web application with error message.

4-2 Leader Home and Dashboard

After a club leader or UVU admins signs in with their credentials, they see a dashboard displaying multiple options for administrators. At the top there are three small summary boxes displaying the upcoming event count, pending hour approvals, and total members in the club. Under the summary boxes the page is split into two columns. On the left column panel is a brief list of upcoming events showing only the event name and date for a quick reference. When the user hovers over and click on an event, they taken to a different page to view or edit the event information. On the right is a list of volunteers who have submitted their hours. Each row has buttons to either approve or deny the hours. Selecting the member will open a new page where admins can view further details. This web interface makes it easy to navigate for leaders to see what needs attention first.

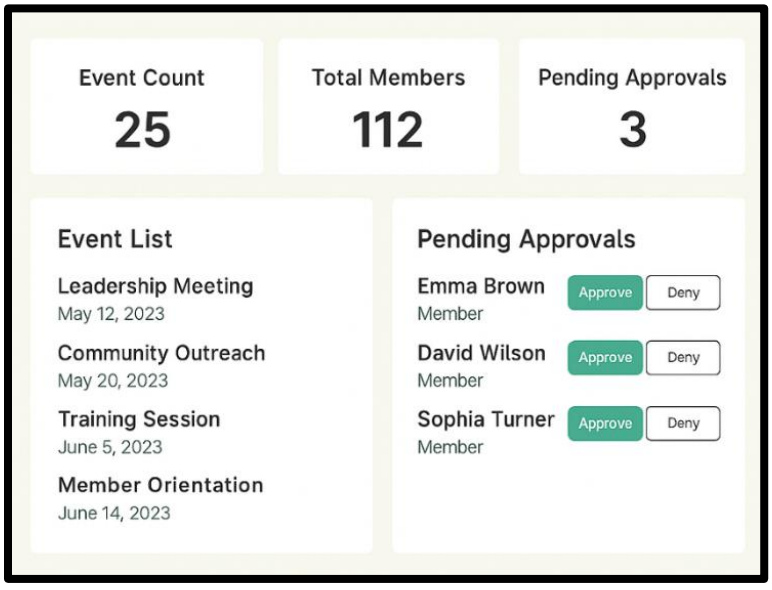


Figure 5 Prototype of Leader Home and Dashboard

4-3 Event Management and Reports

On the Events page, leaders see a table of events with columns for name, date, location, and status such as Open, Full, or Completed. At the bottom right of the Event List section there is a Create Event button. Clicking it opens a form where the leader can enter the event title, date, time, location, description, and how many volunteer spots are available.

In the Reports page, leaders pick a date range like “Fall 2025” and choose how they want to group the data, for example by event or by volunteer. The screen shows totals for hours and participants. On the Reports section there is a Export button that lets club leaders download the report and send it to administrators or the UVU Clubs Office.

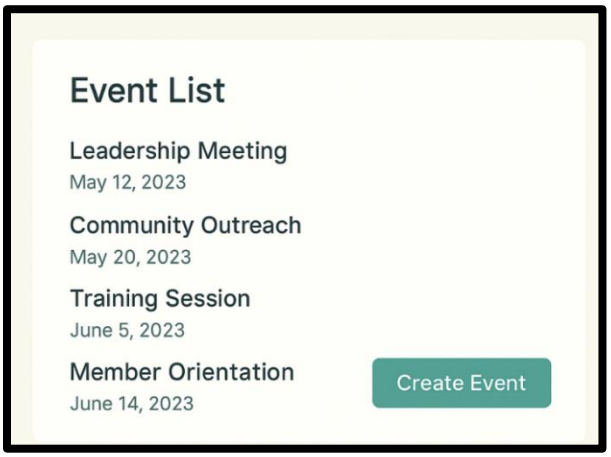


Figure 6 Protype of Event List section.



Figure 7 Protype of Report section.

4-4 Mobile Interface

The mobile layout is mainly focused on the perspective of volunteers and UVU students. Screens are primarily viewed vertically with large text and buttons so they are easy to view and tap. The main navigation has two options, Events and My Hours. When a student opens the app on their phone, they first see a list of upcoming events shown as cards. Each card shows the event name, date and time, and location. There is also a short text displaying "*# of 10 spots filled*" to know if spots are still available.

Tapping a card opens the Event Details screen. This screen displays the basic information with a short description, and a Sign Up button available at the bottom. If the event is already full, the button is disabled and a text pop-up tells the student that there are no spots available. A club could later add a waitlist option in this same area. The My Hours screen lets volunteers see their past and current services. Each row lists the event name, date, number of hours, and a status label such as Pending or Approved. At the bottom of the page is a Submit Hours button. Students can use this when they need to log hours manually if their hours need to be adjusted.



Figure 8 Prototype of Event Details Page

4-5 Help and Error Messages

The error and help messages on the system will be short and clear for users to know what caused the trigger and what to do next. If a user types the wrong email or password on the login page, a message appears under the form that says: "*We could not sign you in. Check your email and password and try again. If you forgot your password, click Forgot password to reset it."* The email and password boxes are outlined red, refer to figure 4, to show exactly where the problem is. If a volunteer or student tries to sign up for an event that just reached its limit, the app blocks the sign up and shows this message: "*This event is full. Please choose another event or check back later."* The Sign Up button is disabled and the student is taken back to the event list where they can pick a different option.



Figure 9 Prototype of Error Message when event availability is full.

5 Appendix Material

Meeting 6 – Written Use Case and Use Case Diagrams

Location: Virtual Microsoft Teams Meeting

Date: 10/24/25

Start Time: 5:00pm

Attendees: Laurent Ortiz, Parker Stewart, Franciso Delarosa

Project Manager assigned tasks to be completed prior to next meeting, 11/7/25. Project Manager and Researcher are tasks to complete the Written and Full Dressed Use Cases for *Create an Event* and *Volunteer Sign Up Event*. System Analyst and Prototype Designer are responsible for developing the *Swim Lane Diagrams* for both events.

Meeting 7 – Finalize Use Cases and Activity Lane Diagrams

Location: Virtual Microsoft Teams Meeting

Date: 11/7/25

Start Time: 5:00pm

Attendees: Laurent Ortiz, Parker Stewart

Project Manager and System Analyst covered for Researchers and Prototype Designer sections on the Use Cases and Activity Lane Diagrams. Researcher could not complete their Use Case tasks due to personal reasons. Prototype Designer has not complete their Use Case tasks due to unknown reasons. We discussed the next steps and how to equally distribute responsibilities for the rest of the members.

Meeting 8 – Process/Logic/Conceptual Modeling and Interface Design

Location: Virtual Microsoft Teams Meeting

Date: 11/14/25

Start Time: 5:00pm

Attendees: Laurent Ortiz, Parker Stewart, Jasper Clark

Project Manager assigned specific tasks for each member to be completed by 12/7/25. The Project Manager is responsible for creating and formatting the final Analysis and Prototype paper and the Process Model section of the paper. System Analyst is responsible for developing the Logic Model and Conceptual Model section of the paper. And the Researcher will be responsible for creating the Interface Design images of the prototype section of the paper. Prototype Designer has not been attending meetings without prior communication and not completing assigned tasks - is no longer part of the team.

Meeting 9 – Draft of Analysis and Prototype Paper

Location: Virtual Microsoft Teams Meeting

Date: 12/5/25

Start Time: 5:00pm

Attendees: Laurent Ortiz, Parker Stewart

The Project Manager and System Analyst check with each other on the development of the ER diagram, data dictionaries, and the final paper. System Analyst fixed the structure of the ER diagram for it to be able to format onto the final paper. Project Manager will reach out to Researcher to check in on their progress and if they need any assistance.

Meeting 10 – Finalize Analysis and Prototype Presentation

Location: Virtual Microsoft Teams Meeting

Date: 12/7/25

Start Time: 5:00pm

Attendees: Laurent Ortiz, Parker Stewart, Jasper Clark

All members collaborated on preparing for the Analysis & Design presentation. Each member focused on adding information on their assigned tasks. After completing the slides, we recorded ourselves presenting the presentation. The project manager advised the rest of the members she will review the final paper and submit the final deliverables by Wednesday or sooner.